UNIT 3 · TOPIC 1(B)

Calibration of Planters

Syllabus topic: Calibration of seed drills, planters and plant protection equipment

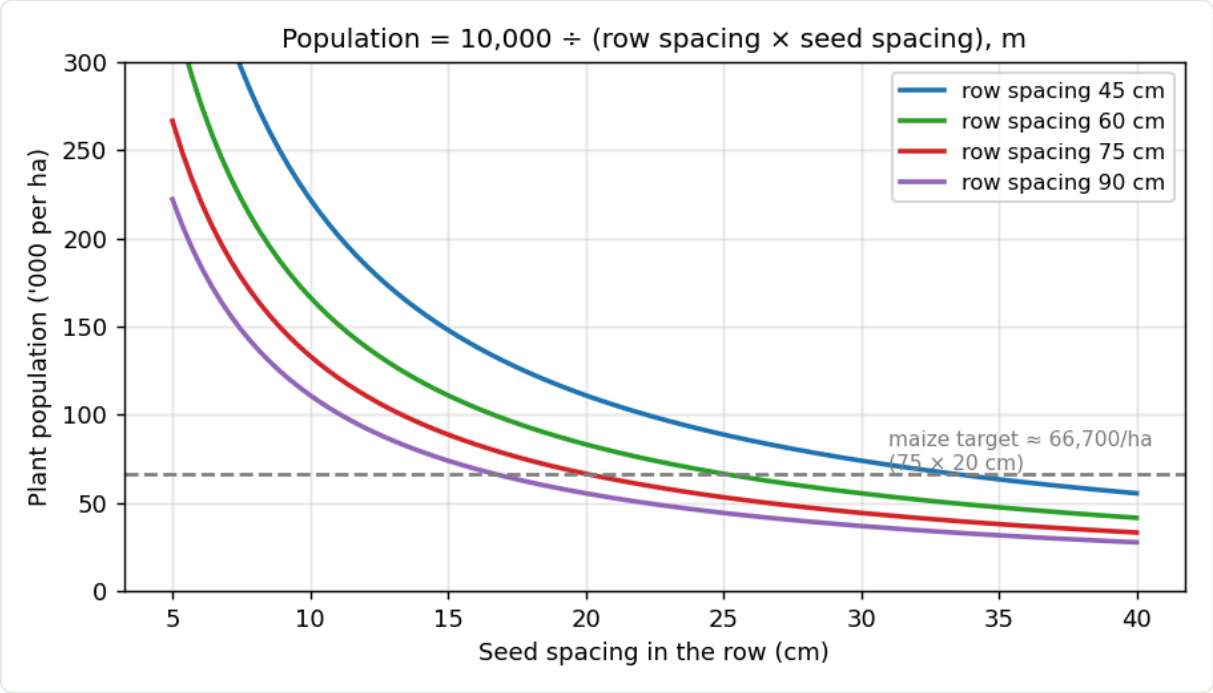
Precision (single-seed) planters

Precision planters (horizontal/vertical/inclined plate, or vacuum disc) meter ONE seed per cell.

$$\text{Seed spacing } x = \pi D \div (K \times n)$$

D = drive wheel diameter; n = number of cells on the plate; K = plate-to-wheel transmission (speed) ratio. Adjust spacing by changing sprockets (K) or plates (n).

$$\text{Plant population (plants/ha)} = 10{,}000 \div [\text{row spacing (m)} \times \text{seed spacing (m)}]$$



Plant population vs seed spacing for different row spacings

Stationary check

1. Rotate the wheel N revolutions.
2. Catch the seeds; count and weigh them. Expected seeds = **N × n × K**.
3. Compare actual vs expected.
4. Check spacing statistics on a greased belt/sticky tape (lab), or by uncovering seeds in a field strip.

Precision-planting indices (target spacing x)

Index	Definition
MISS index	% of intervals > 1.5x (cell failed)
MULTIPLE index	% of intervals < 0.5x (doubles)
QUALITY OF FEED index	% of intervals between 0.5x and 1.5x (singles — higher is better)
Precision (C3)	CV of the single intervals

The same indices are used in planter testing (Topic 2c).

Vacuum planters

- Check plate rpm vs forward speed, vacuum level (seed-size dependent) and singulator setting.
- Calibrate the seed rate by **count, not weight**, for costly hybrid seed.

Practice MCQs

Q12. Seed spacing of a plate planter with wheel diameter D, cell number n and plate/wheel speed ratio K is:

(a) $\pi D/(K \cdot n)$ (b) $K \cdot n/\pi D$
(c) $\pi DK/n$ (d) $n/(\pi DK)$

Q15. In precision-planting evaluation, an interval greater than 1.5× target spacing counts in the:

(a) Multiple index (b) Miss index
(c) Quality of feed (d) Precision

Q16. An interval less than 0.5× target spacing indicates:

(a) A miss (b) A double/multiple
(c) A single (d) Slip

Q17. Quality of feed index is the percentage of intervals:

(a) >1.5x (b) <0.5x
(c) Between 0.5x and 1.5x (d) Exactly x

Q18. In vacuum planters, seed singulation is achieved by:

(a) Gravity (b) Suction holding one seed per hole with a singulator
(c) Springs (d) Brushes only

Q34. Expected seeds dropped in N wheel revolutions from a plate with n cells and speed ratio K:

(a) $N \cdot n \cdot K$ (b) $N/(nK)$
(c) πDN (d) nK/N

Answers

Q12 — (a) $\pi D/(K \cdot n)$
Q15 — (b) Miss index
Q16 — (b) A double/multiple
Q17 — (c) Between 0.5x and 1.5x
Q18 — (b) Suction holding one seed per hole with a singulator
Q34 — (a) $N \cdot n \cdot K$

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting) & Ch. 21 (Spraying and Dusting); Jagdishwar Sahay — Elements of Agricultural Engineering; ISI/BIS & RNAM test codes for seed drills and sprayers; NPTEL Farm Machinery lectures.
